Qi Sun

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Research Interests: Efficient and reliable language-model agents, with a focus on on-policy distillation, model merging, and retrieval for clinical applications.

EDUCATION

New York University Shanghai Shanghai, China
B.S. in Computer Science Expected May 2027
Independent capstone: *Right Finding, Wrong Organ: Auditing Anatomical Scope in Clinical Report Retrieval*

PUBLICATIONS & PREPRINTS

[1] **Qi Sun**[*], Siyue Zhang[*†], Yulin Chen, Yuxiang Xue, Ru Peng, Chen Zhao[†]. "From 'Weak' Signals to Strong Models: Preference Delta Aggregation with LoRA Merging." **NeurIPS 2026.** arXiv:2606.00357.

[2] Yizhou Liu[*], **Qi Sun**[*], Yulin Chen, Siyue Zhang, Chen Zhao. "Search, Do not Guess: Teaching Small Language Models to Be Effective Search Agents." *Committed to EACL 2027 (decision pending)*. arXiv:2604.04651.

[3] **Qi Sun**[*], Siyue Zhang[*†], Tingyu Song, Yilun Zhao, Chen Zhao[†]. "Rethinking Offline On-Policy Distillation: From Scalar Advantages to Distributional Supervision." *Under review at ACL Rolling Review (October 2026)*.

[*]Equal contribution. [†]Corresponding author.

RESEARCH EXPERIENCE

Undergraduate Researcher 2024.06 – Present
Advisor: Prof. Chen Zhao NYU Shanghai

Distributional on-policy distillation for agentic search (DistOPD) [3] 2026.06 – Present

- Found that offline on-policy distillation (Lightning OPD) improves math reasoning but consistently falls below its SFT starting point in multi-turn agentic search, and traced the failure, empirically and theoretically, to distributional flattening under scalar-advantage supervision on fixed rollouts.
- Developed DistOPD, an offline method that keeps teacher and reference probabilities over a small set of candidate tokens plus a residual token, so the teacher scores each rollout once and is never queried during training.
- Matched online OPD on agentic search with 10.6× fewer training GPU-hours. Led the failure analysis, method design, and experiments.

Preference Delta Aggregation for weak-to-strong training (PDA) [1] 2026.01 – 2026.05

- Proposed PDA, which turns each pair of a weak and a weaker model (e.g., Qwen3-4B over 1.7B) into a preference delta trained as a LoRA adapter, and aggregates many deltas by LoRA merging, with Geometric Alignment Merging (GAM) to reduce interference between adapters.
- Improved Qwen3-8B by 6.8 points on knowledge reasoning and 7.3 points on agentic search (benchmark averages), 2.1 and 4.3 points above the best single-delta baseline. Presenting author at NeurIPS 2026.

Small language models as effective search agents [2] 2025.06 – 2025.12

- Reproduced Search-o1 and Search-R1 as controlled baselines, distilled agentic search from Qwen3-32B into Qwen3-1.7B with SFT and on-policy distillation, and designed the probing and rejection fine-tuning experiments on when to search.
- Found that distilled small models search less and hallucinate more, and that adaptive when-to-search strategies often hurt them. Training them to search consistently and ground answers in retrieved evidence improves over agent distillation by 17.3 points on Bamboogle and 15.3 on HotpotQA.
- Supported by the Dean's Undergraduate Summer Research Fund.

Research Assistant 2026.05 – Present
Advisor: Prof. Yiqiu (Artie) Shen NYU Langone Health, New York, NY

Right Finding, Wrong Organ: Auditing Anatomical Scope in Clinical Report Retrieval

- Built a feature-oriented retrieval system over abdominal CT/MRI reports and structured EHR that surfaces similar cases for ambiguous pancreatic presentations, using a clinical codebook that encodes findings, quantitative thresholds, and negation for hybrid BM25 and dense retrieval.
- Audited whether retrieved evidence comes from the right anatomical scope, a failure in which a retriever matches the right finding in the wrong organ. This work is also my independent capstone. Manuscript in preparation for a medical informatics journal.

HONORS & AWARDS

- Dean's Undergraduate Summer Research Fund, NYU Shanghai 2025
- ICPC Regional Contest (Ningxia), Bronze Prize 2025
- CSAW LLM CTF 2024 (MENA), Second Place 2024

LEADERSHIP

- **Chair,** NYU Shanghai ACM Student Chapter 2024.06 – 2025.06
- **Director of Competition,** NYU Shanghai ICPC Training Community 2024.06 – 2025.06
  With Prof. Xianbin Gu, built the team's training platform (Vue3, Django) with an online judge and LLM-based personalized problem recommendation.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, SQL, JavaScript
- **ML Stack:** PyTorch, Hugging Face Transformers, PEFT/LoRA, vLLM, FAISS, Slurm multi-GPU clusters